

Genome-wide analysis of a land plant-specific *acyl:coenzymeA synthetase* (ACS) gene family in *Arabidopsis*, poplar, rice and *Physcomitrella*

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Summary

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- The plant enzyme 4-coumarate:coenzyme A ligase (4CL) is part of a family of adenylate-forming enzymes present in all organisms. Analysis of genome sequences shows the presence of '4CL-like' enzymes in plants and other organisms, but their evolutionary relationships and functions remain largely unknown.

- 4CL and 4CL-like genes were identified by BLAST searches in *Arabidopsis*, *Populus*, rice, *Physcomitrella*, *Chlamydomonas* and microbial genomes. Evolutionary relationships were inferred by phylogenetic analysis of aligned amino acid sequences. Expression patterns of a conserved set of *Arabidopsis* and poplar 4CL-like *acyl-CoA synthetase* (ACS) genes were assayed.

- The conserved ACS genes form a land plant-specific class. Angiosperm ACS genes grouped into five clades, each of which contained representatives in three fully sequenced genomes. Expression analysis revealed conserved developmental and stress-induced expression patterns of *Arabidopsis* and poplar genes in some clades.

- Evolution of plant ACS enzymes occurred early in land plants. Differential gene expansion of angiosperm ACS clades has occurred in some lineages. Evolutionary and gene expression data, combined with *in vitro* and limited *in vivo* protein function data, suggest that angiosperm ACS enzymes play conserved roles in octadecanoid and fatty acid metabolism, and play roles in organ development, for example in anthers.

Key words: 4CL-like, 4-coumarate:CoA ligase, *acyl-CoA synthetase*, *acyl activating enzyme* (AAE), adenylate forming enzyme, genome sequences, jasmonic acid, land plant evolution.

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Introduction

The enzyme 4-coumarate:CoA ligase (4CL) plays important roles in phenylpropanoid metabolism by generating CoA esters of hydroxycinnamic acids. These cinnamyl CoA esters are used as intermediates in the biosynthesis of a large array of phenolic secondary natural products, including monolignols and flavonoids (Hahlbrock & Scheel, 1989). The first 4CL gene was cloned from parsley (Douglas *et al.*, 1987), and 4CL

enzymes are encoded by multigene families in all vascular plants examined to date (Lee & Douglas, 1996; Allina *et al.*, 1998; Hu *et al.*, 1998; Ehrling *et al.*, 1999; Lindermayr *et al.*, 2002; Kumar & Ellis, 2003; Hamberger & Hahlbrock, 2004; Tuskan *et al.*, 2006; Hamberger *et al.*, 2007). Analysis of enzymatic properties of recombinant enzymes has revealed that 4CL isoenzymes have differential activity towards different hydroxycinnamyl substrates (Lee & Douglas, 1996; Allina *et al.*, 1998; Hu *et al.*, 1998; Ehrling *et al.*, 1999;

Stuible & Kombrink, 2001; Lindermayr *et al.*, 2002; Hamberger & Hahlbrock, 2004). The analysis of the *4CL* gene family in the fully sequenced *Arabidopsis* (Ehlting *et al.*, 1999; Hamberger & Hahlbrock, 2004), poplar (Tsai *et al.*, 2006; Tuskan *et al.*, 2006; Hamberger *et al.*, 2007) and rice (Tuskan *et al.*, 2006; Hamberger *et al.*, 2007) genomes showed that *4CL* is encoded by four to five genes in these plants. Differential *4CL* gene expression patterns in *Arabidopsis* and poplar, coupled with *4CL* isoenzyme substrate utilization preferences, suggest that *4CL* genes and enzymes have undergone subfunctionalization for the biosynthesis of different classes of phenylpropanoid-derived compounds (Ehlting *et al.*, 1999; Harding *et al.*, 2002), with the phylogenetically distinct Class I and Class II clades of *4CL*s specialized for monolignols and flavonoid biosynthesis, respectively.

The *4CL* enzymes are members of the adenylate-forming enzyme superfamily, which share a common reaction involving formation of an adenylate intermediate (Becker-Andre *et al.*, 1991; Schneider *et al.*, 2005), including those involved in fatty acid chain elongation (Shockey *et al.*, 2003). Following the generation of genome sequence data from *Arabidopsis*, a number of genes encoding adenylate-forming enzymes were annotated as being closely related to true *4CL*s but of unknown specific biochemical function, and these may function in diverse pathways in plant metabolism and natural product biosynthesis. For example, initial *Arabidopsis* genome sequence data revealed the presence of four genes with distant similarity to *4CL* genes (Cukovic *et al.*, 2001). In a subsequent analysis, eight members of a larger set of *Arabidopsis* *4CL*-like adenylate-forming enzymes annotated in the completed *Arabidopsis* genome were classified as *4CL*-like genes because of their close phylogenetic relationship to *bona fide 4CL*s (Raes *et al.*, 2003), with which they share structural similarities such as a conserved substrate binding domain and conserved Box I and II domains (Ehlting *et al.*, 2001; Schneider *et al.*, 2003). As summarized by Ehlting *et al.* (2005), a total of nine genes encoding enzymes most closely related to genes encoding *4CL* have been annotated as *4CL* or *4CL*-like by different groups (Costa *et al.*, 2003; Raes *et al.*, 2003; Shockey *et al.*, 2003). Interestingly, in contrast to *4CL* proteins, most *4CL*-like enzymes are predicted to be targeted to the peroxisome because of the presence of C-terminal peroxisome targeting sequence 1 (PTS1) sequences, and such localization has been experimentally verified in three cases (Schneider *et al.*, 2005; Koo *et al.*, 2006). According to the analysis of Reumann *et al.* (2004) nine major PTS1 tripeptides ([SA][RK][LM]> without AKM> plus SRI> and PRL>) located at the C-terminus of plant proteins are strong indicators for peroxisomal localization, with SRL, SRM, SKL, ARL and PRL being among the most common sequences, while other less frequently observed sequences such as SNL are also functional.

Unlike true *4CL* genes, expression of *4CL*-like genes is not associated with lignification or flavonoid biosynthesis (Raes *et al.*, 2003; Ehlting *et al.*, 2005), suggesting that they

may not encode enzymes with *4CL* activity and lack activity towards known *4CL* hydroxycinnamate substrates (*p*-coumaric, caffeic, ferulic, 5-hydroxyferulic and sinapic acids). Indeed, *in vitro* testing of seven *4CL*-like recombinant proteins (At5g63380, At4g05160, At4g19010, At3g48990, At1g20510, At1g62940 and At5g38120) with phenylpropanoid pathway intermediates did not result in any measurable catalytic activity (Costa *et al.*, 2005). Independently, the enzymes encoded by several *4CL*-like genes were shown to be acyl-coenzyme A synthetases that accept medium to long-chain fatty acids and in some cases the cyclopentenone 12-oxo-phytodienoic acid (OPDA) and/or OPDA derivatives as *in vitro* substrates (Schneider *et al.*, 2005; Kienow *et al.*, 2008). OPDA is an intermediate in the octadecanoid pathway leading jasmonic acid (JA) biosynthesis. The later steps of this pathway, formation of OPDA-CoA thioesters followed by beta-oxidation of OPDA-CoA resulting in acyl chain shortening, occur in the peroxisome (Li *et al.*, 2005; Koo *et al.*, 2006). Recently, the *Arabidopsis 4CL*-like gene At1g20510 has been shown to encode a peroxisomal OPDA:CoA ligase involved in JA biosynthesis, and is now designated *OPCL1* (Koo *et al.*, 2006; Kienow *et al.*, 2008). However, the biological functions of remainder of the *Arabidopsis* genes originally annotated as *4CL*-like are still unknown. Based on the activities of several *4CL*-like enzymes as acyl-CoA synthetases, and their lack of activity against hydroxycinnamate substrates, we use the name *acyl-CoA synthetase (ACS)* for the remainder of the *4CL*-like genes in *Arabidopsis* without known *in vivo* functions, and for homologues of these genes in other plants. Completion and annotation of the rice (Yuan *et al.*, 2005; <http://www.tigr.org/tdb/e2k1/osa1/>), poplar (Tuskan *et al.*, 2006; http://genome.jgi-psf.org/Poptr1_1/Poptr1_1.home.html), and *Physcomitrella* (http://genome.jgi-psf.org/Phypa1_1/Phypa1_1.home.html) genomes opens the door to comparative genomics approaches aimed at understanding the evolution and potential functions of genes such as those in the *4CL*-like *ACS* family, using *Arabidopsis* as a reference genome (*Arabidopsis* Genome Initiative, 2000; <http://www.arabidopsis.org/>).

In this study, we used *Arabidopsis 4CL* sequences for BLAST searches of eukaryotic and prokaryotic genome databases to identify over 100 *4CL*-related genes in diverse taxa, and identified and annotated *ACS* and genes encoding related adenylate-forming enzymes in the fully sequenced poplar, rice, and *Physcomitrella* genomes. In addition, we retrieved full-length *ACS* sequences from a maize genome database and publicly available plant genome databases. Phylogenetic reconstructions based on amino acid sequence alignments showed that *ACS* genes, including the nine canonical *Arabidopsis 4CL*-like *ACS* genes, belong to a land plant-specific clade most closely related to true *4CL*s, and that each fully sequenced angiosperm plant genome has representatives in each of the five well-defined *ACS* clades, four of which contain proteins predominantly predicted to be peroxisomally localized.

This suggests that ACS enzymes perform important, conserved roles in land plant metabolism. We profiled the developmental and stress-induced expression of *Arabidopsis* ACS genes and selected poplar homologues, representing all five clades. The combination of phylogenetic reconstruction, gene expression, and *in silico* co-expression analyses provided insights into the evolutionary diversification of ACS function in angiosperm lineages, subfunctionalization of duplicated ACS genes, identification of putative poplar orthologues of *Arabidopsis* ACS genes, and potential functions of representative ACS genes.

Materials and Methods

Plant growth conditions

Arabidopsis thaliana (*Arabidopsis*) seedlings were transferred from MS agar plates to soil (Sunshine mix 5; Sungrow Horticulture, Saba Beach, Alberta, Canada) and cultivated at 20°C under long-day conditions (18 h light) until maturity. Poplar (*Populus trichocarpa* × *Populus deltoides* genotype H11-11) plants were grown indoors at the University of British Columbia Horticulture Greenhouse, Vancouver, Canada, as described in Ralph *et al.* (2006).

Gene expression analysis

Arabidopsis total RNA was isolated from the specified tissues frozen in liquid nitrogen, ground to a fine powder, and RNA extracted using the Trizol reagent (Gibco BRL, Grand Island, NY, USA) following manufacturer's instructions. The quality of the RNA samples was assessed by visual inspection of the rRNA bands on a 1% agarose gel. RNA samples were quantified spectrophotometrically and 2 µg DNase I-treated RNA per 20 µl reaction was used to generate cDNA using Superscript II Reverse Transcriptase (Invitrogen, Carlsbad, CA, USA) following the manufacturer's protocol. Poplar RNA was extracted from various poplar organs and cDNA prepared as described (Ralph *et al.*, 2006).

For the quantitative reverse transcriptase polymerase chain reaction (qRT-PCR) analysis of *Arabidopsis* gene expression, 10 ng of cDNA was incubated with 10 µl QuantiTect SYBR Green PCR mastermix (Qiagen Inc., Valencia, CA, USA) and 30 nmol of each a forward and a reverse primer in a total volume of 20 µl. After an initial denaturation step at 95°C for 15 min, 40 cycles at 95°C for 15 s, 55°C for 30 s and 68°C for 45 s were followed by a fluorescence reading. After a final incubation at 68°C for 5 min, a melting curve was generated ranging from 90°C to 60°C. Threshold cycles (CT) were adjusted manually, and the CT values for a housekeeping control amplified in parallel on each plate were subtracted from CT values obtained for each gene of interest, thus generating normalized CT values (Δ CT). The relative starting quantities of each gene were determined by setting as a base value the gene with the highest CT value within a tissue panel or

treatment series, and relative quantities were calculated using the $\Delta\Delta$ CT method, as described in Hietala *et al.* (2003). The Δ CT values were calculated after normalization using the following control genes: *Arabidopsis* adenine phosphoribosyl-transferase (*APT1* – At1g27450) for all *Arabidopsis* expression experiments and poplar eukaryotic translation initiation factor 5A-1/eIF-5A-1 (GenBank accession CV251327; Ralph *et al.*, 2006) for all poplar experiments. The $\Delta\Delta$ CT value was calculated using the following reference tissues: the highest expressing tissue for the developmental expression panel (see Fig. 3); and unwounded leaf tissues in the stress treatment panels (see Figs 4 and 5). Intron-spanning primers used to amplify each gene are given the Supporting Information, Table S1. Selected qRT-PCR reaction products were sequenced to insure fidelity of the amplifications.

Recombinant DNA methods

Plasmid DNA was prepared using Qiagen spin Miniprep and Midiprep kits, following the manufacturer's instructions. DNA sequencing was performed by the University of British Columbia Nucleic Acid and Protein Service unit, using BigDye 3.0 and a Prism Sequencer (Applied Biosystems, Foster City, CA, USA).

Arabidopsis ACS promoter regions were amplified from genomic DNA using the *pwo* polymerase (Roche) and cloned into the *Nco*I site of the pCambia 1305.1 vector (www.cambia.org), preserving the start codon of the *GUS* (beta-glucuronidase) reporter gene (Jefferson *et al.*, 1987). Primers containing the compatible restriction site sequence are given in the Supporting Information, Table S2.

The coding sequences of both *AtOPCL1* and *PoptrACS5* were polymerase chain reaction (PCR) amplified using cDNA derived from organs where these genes were most highly expressed using Phusion high fidelity enzyme (Finnzymes, Espoo, Finland) according to the manufacturer's protocol. The PCR primers are given in the Supporting Information, Table S3. The PCR products were cloned into a Gateway (Invitrogen) compatible entry vector using TOPO-TA cloning kit (Invitrogen) and subsequently recombined into the destination vector using LR Clonase II enzyme mix according to the manufacturer's instructions (Invitrogen). The PCR products were cloned in-frame *N*-terminal to the *GFP* gene driven by the CaMV35S promoter.

Bioinformatic and phylogenetic reconstruction methods

The set of *Arabidopsis* genes encoding 4CL enzymes (Ehltng *et al.*, 1999) was used in BLAST homology searches to identify potential 4CL-like/ACS genes in the *Arabidopsis* genome, using the database maintained at The *Arabidopsis* Information Resource (<http://www.arabidopsis.org>). Poplar homologues were identified by reciprocal BLAST searches of the poplar

genome assembly (Joint Genome Institute, *Populus trichocarpa* v.1.1; http://genome.jgi-psf.org/Poptr1_1/Poptr1_1.home.html) using *At4CL* and *AtACS* sequences as queries. The poplar gene models (from automated *ab initio* gene-calling programs; Tuskan *et al.*, 2006) assigned for a given locus were evaluated, annotated manually and revised as necessary (Table 1 and the Supporting Information, Table S4). All annotated candidates corresponded to loci anchored to poplar linkage groups or to sequence scaffolds, as described in Tuskan *et al.* (2006). Corresponding rice homologues (Table 1 and Table S4) were identified in the rice genome using BLAST searches of the rice genome annotation at The Institute for Genome Research (TIGR; <http://www.tigr.org/tdb/e2k1/osa1/>). *Physcomitrella* sequences were identified using the JGI website (http://genome.jgi-psf.org/Phypa1_1/Phypa1_1.home.html). All sequences are given in Table 1 and Table S4. Selected microorganism sequences were obtained by BLAST searches using *At4CL* and *AtACS* sequences as queries at the NCBI website (<http://www.ncbi.nlm.nih.gov/blast/Blast.cgi>). The accession numbers of these sequences are given in Table S4. Protein sequences were aligned using the Genomatix Dialign program (<http://www.genomatix.de/cgi-bin/dialign/dialign.pl>) and the multiple protein sequence alignments were manually optimized. To reconstruct phylogenetic trees, maximum likelihood analyses with 100 bootstrap replicates were carried out using PHYL v. 2.4.4 (Guindon & Gascuel, 2003) with the JTT model of amino acid substitution.

β -Glucuronidase (GUS) and green fluorescent protein (GFP) report gene activity

To generate transgenic *Arabidopsis* Col-0 plants expressing the *GUS* reporter under control of *ACS* gene promoters, plants were transformed following the floral dip protocol (Clough & Bent, 1998). Five to ten transgenic lines per construct were surveyed for histochemical GUS activity at various stages of development, and wound-induced GUS activity was assayed in representative lines.

The GUS histochemical assay solution was prepared by mixing an aqueous solution of 100 mM Na_3PO_4 , pH 7.0, 0.5% X-GLUC (bromochloroindolyl-b-glucuronide) with an aqueous solution of 2 mM $\text{K}_3\text{Fe}(\text{CN})_6$ and 2 mM $\text{K}_4\text{Fe}(\text{CN})_6$ in 0.1% Triton X. Young *Arabidopsis* leaf blades were wounded with scissors, cut from the plant after 1 h and placed in cold GUS assay solution, vacuum infiltrated for 15 min and incubated at 37°C for 2 h or until a blue colour could be observed. Chlorophyll was removed from the tissue by incubation in 95% ethanol overnight. Results were documented using a Leica dissecting microscope equipped with a Spot32 camera at the UBC BioImaging Facility, Vancouver, Canada.

Agrobacterium strains carrying *GFP::ArathACS4* and *GFP::PoptrACS5* were used to generate eight independent transgenic tobacco lines by leaf disc transformation (Hauffe

et al., 1991) and plants were screened for fluorescence indicating *GFP* expression under an epifluorescence microscope (Zeiss Axioplan 2). Plants with both high and low levels of GFP fluorescence were selected for analysis by confocal microscopy (Zeiss Meta Confocal). Wild-type tobacco plants were used as negative controls. Chloroplast autofluorescence was excited with a 488 nm argon laser and detected after passage through a long-pass 650 nm emission filter. The GFP fluorescence was excited with a 488 nm laser and detected after passage through a band pass 505–530 nm emission filter. Images were reconstructed using the IMAGEJ software suite (Rasband, 1997–2007).

Results and Discussion

Genome-wide phylogenetic analysis Acyl:CoA synthetase related genes

The adenylate-forming enzyme superfamily that uses a common reaction mechanism includes members from all organisms studied to date, including prokaryotes and eukaryotes (Becker-Andre *et al.*, 1991; Shockey *et al.*, 2003; Schneider *et al.*, 2005). We identified 88 full-length genes encoding adenylate-forming enzymes related to 4CL from genomic databases, using *in silico* similarity searches based on the amino acid sequences of *Arabidopsis* 4CL proteins (Ehltig *et al.*, 1999; Hamberger & Hahlbrock, 2004). In this analysis, we focused on three angiosperms with complete genome sequences available (*Arabidopsis*, poplar and rice), the genomes of maize, *Physcomitrella*, *Chlamydomonas* and the genomes of selected other microorganisms (fungi and bacteria) for which complete or substantial genome sequence data is available.

A phylogenetic analysis of over 100 sequences from these various organisms, including *bona fide* 4CL sequences from *Arabidopsis*, poplar, and rice, is shown in Fig. 1. This analysis revealed two general groups of adenylate-forming proteins. One large group contained representatives from all organisms analysed, including bacteria, fungi, *Chlamydomonas*, *Physcomitrella* and angiosperm plants (large arc in Fig. 1). These probably represent adenylate-forming enzymes with metabolic functions that are conserved in many lineages. As an example of possible functions, one clade in this group contains the *Arabidopsis* *ACN1* gene, which encodes an acetate:CoA ligase that functions as an entry point to the glyoxylate cycle during seed germination (Turner *et al.*, 2005). The clade containing *ACN1* is enriched for angiosperm plant sequences, and includes *Arabidopsis* genes encoding the acyl activating enzymes (AAE) AAE4 and AAE6, which may play housekeeping functions related to fatty acid metabolism (Shockey *et al.*, 2003), as well as two *Chlamydomonas* genes. A sister clade contains a *Saccharomyces cerevisiae* gene encoding the FAT2 peroxisomal acyl-CoA synthetase, as well as the *Arabidopsis* AAE3 gene and several poplar, rice, and *Physcomitrella* genes, all of unknown

Table 1 Poplar, rice, *Physcomitrella* and *Arabidopsis* 4CL (4-coumarate:coenzyme A ligase) and ACS (acyl-CoA synthetase) genes

Name	Clade ¹	Gene model	Predicted PTS1 targeting sequence ²
Poplar			
<i>Poptr4CL1</i>	4CL	estExt_fgenes4_pg.C_1210004 scaffold ³	–
<i>Poptr4CL2</i>	4CL	gw1.XVIII.2818.1 LG_XVIII	–
<i>Poptr4CL3</i>	4CL	grail3.0100002702 LG_I	–
<i>Poptr4CL4</i>	4CL	grail3.0099003002 LG_IX	–
<i>Poptr4CL5</i>	4CL	fgenes4_pg.C_LG_III001773 LG_III	–
<i>PoptrACS1</i>	B	eugene3.01230068	SKM
<i>PoptrACS2</i>	B	estEXT_fgenes1_pg_v1.C_LG_IV0024	SKL
<i>PoptrACS3</i>	C	fgenes4_pg.C_LG_III000782	SKL
<i>PoptrACS4</i>	D	eugene3.00020113	SKL
<i>PoptrACS5</i>	D	fgenes4_pm.C_LG_V000686	SKL
<i>PoptrACS6</i>	E	fgenes4_pm.C_LG_X000932	–
<i>PoptrACS7</i>	E	fgenes4_pm.C_LG_VIII000094	SKL
<i>PoptrACS8</i>	E	eugene3.00640074	SKM
<i>PoptrACS9</i>	E	fgenes4_pm.C_LG_X000174	SKL
<i>PoptrACS10</i>	E	estEXT_fgenes4_pm.C_LG_XV0272	SKL
<i>PoptrACS11</i>	E	eugene3.00120875	SKL
<i>PoptrACS12</i>	E	grail3.0015024001	–
<i>PoptrACS13</i>	A	eugene3.00010460	–
Rice			
<i>Os4CL1</i>	4CL	Os08g14760 ⁴	–
<i>Os4CL2</i>	4CL	Os02g46970	–
<i>Os4CL3</i>	4CL	Os02g08100	–
<i>Os4CL4</i>	4CL	Os06g44620	–
<i>Os4CL5</i>	4CL	Os08g34790	–
<i>OsACS1</i>	B	Os03g05780	SKL
<i>OsACS2</i>	C	Os10g42800	SRL
<i>OsACS3</i>	C	Os08g04770	SKL
<i>OsACS4</i>	D	Os03g04000	SKL
<i>OsACS5</i>	E	Os01g67530	SKL
<i>OsACS6</i>	E	Os01g67540	SRL
<i>OsACS7</i>	E	Os07g17970	SRL
<i>OsACS8</i>	N/A ⁵	Os07g44560	SRM
<i>OsACS9</i>	A	Os04g24530	–
<i>Physcomitrella</i>			
<i>Pp4CL1</i>	4CL	fgenes1_pg.scaffold_185000.I.045 ⁶	–
<i>Pp4CL2</i>	4CL	e_gw1.167.67.1	–
<i>Pp4CL3</i>	4CL	estExt_fgenes1_pm.C_200030	–
<i>Pp4CL4</i>	4CL	estExt_fgenes2_pg.C_710053	–
<i>PpACS1</i>	Physco	estExt_Genewise1.C_520075	SNL
<i>PpACS2</i>	Physco	e_gw1.6.71	PRL
<i>PpACS3</i>	Physco	e_gw1.12.278.1	SKL
<i>PpACS4</i>	Physco	e_gw1.38.65.1	SNL
<i>PpACS5</i>	N/A ⁴	e_gw1.356.14.1	–
<i>PpACS6</i>	A	fgenes1_pg.scaffold_96000119	–
<i>Arabidopsis</i>			
<i>At4CL1</i>	4CL	At1g51680/4CL1 ⁷	–
<i>At4CL2</i>	4CL	At3g21240/4CL2	–
<i>At4CL3</i>	4CL	At1g65060/4CL3	–
<i>At4CL4</i>	4CL	At3g21230/4CL4	–
<i>AtACS1</i>	D	At1g20480/4CL-like4	SKL
<i>AtACS2</i>	D	At1g20490/4CL-like3	SKL
<i>AtACS3</i>	D	At1g20500/4CL-like2	SKL
<i>AtOPCL1</i>	D	At1g20510/4CL-like1	SKL
<i>AtACS5</i>	A	At1g62940/4CL-like5	–
<i>AtACS6</i>	B	At4g05160/4CL-like7	SKM
<i>AtACS7</i>	C	At4g19010/4CL-like6	SRL
<i>AtACS8</i>	D	At5g38120/4CL-like9	SKI
<i>AtACS9</i>	E	At5g63380/4CL-like8	SKL

¹Clades defined in Figs 1 and 2.²PTS1 (peroxisome targeting sequence 1) C-terminal tripeptide peroxisome targeting sequence, if present, based on Reumann *et al.* (2004).³Poplar gene models at the JGI *Populus trichocarpa* v.1.1 genome browser; http://genome.jgi-psf.org/Poptr1_1/Poptr1_1.home.html; 4CL gene models previously published in Tuskan *et al.* (2006); Hamberger *et al.* (2007).⁴Rice gene models at The Institute for Genome Research, TIGR. 4CL gene models previously published in Hamberger *et al.* (2007).⁵N/A, not applicable. Do not fall into delineated clades in Figs 1 and 2.⁶*Physcomitrella* gene models at the JGI *Physcomitrella patens* ssp *patens* v.1.1 genome browser; http://genome.jgi-psf.org/Phypa1_1/Phypa1_1.home.html⁷*Arabidopsis* AGI code and gene name in *Arabidopsis* Monoglignol Biosynthesis Gene Families at The *Arabidopsis* Information Resource (TAIR; <http://www.arabidopsis.org/browse/genefamily/Raes.jsp>)

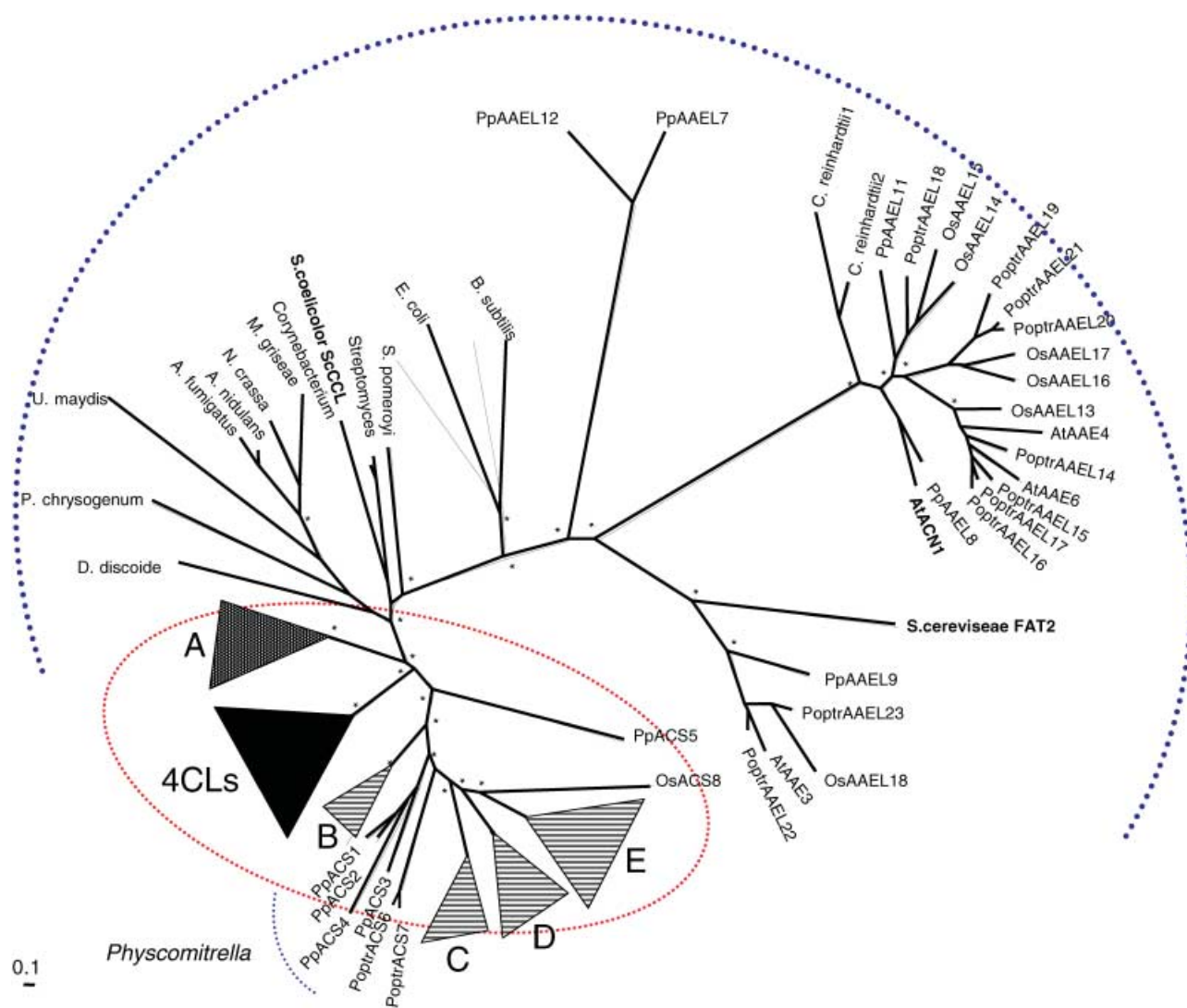


Fig. 1 Phylogenetic tree of 104 proteins related to 4-coumarate:CoA ligase (4CL). 4CL-related (acyl-CoA synthetase (ACS)) sequences corresponding to translated nucleotide sequences from full-length cDNAs and expressed sequence tags (ESTs) were aligned, and an unrooted phylogenetic tree was generated. Nodes with bootstrap values above 70% are indicated by asterisks. Triangles represent the subclades of land plant-specific ACSs most closely related to true 4CL enzymes. The large dashed arc indicates a large group of genes found in all organisms from all lineages investigated (land plants, algae, fungi, protists and prokaryotes); the dashed oval indicates the clade of land plant-specific ACS genes, as discussed in the text. Solid triangle, true 4CL enzymes; stippled triangle, nonperoxisomal ACSs in clade A; hatched triangles, ACS clades B–E as described in the text. A *Physcomitrella* clade of land-plant ACSs is indicated. Gene name prefixes: At, *Arabidopsis thaliana*; Os, *Oryza sativa*; Poptr, *Populus trichocarpa*; Pp, *Physcomitrella patens*. Gene names and identifiers are given in Table 1 and the Supporting Information, Table S4. The scale represents 0.1 amino acid changes.

function. The ACS protein encoded by the bacterium *Streptomyces coelicolor* A3(2), ScCCL, has been shown to have high activity against 4-coumaric and cinnamic acids and is thus designated as a cinnamic acid-CoA ligase, ScCLL (Kaneko *et al.*, 2003). This raises the possibility that other bacterial and fungal enzymes in this clade have activities against phenolic substrates. Based on their similarity to *Arabidopsis* AAE genes, we designated the annotated poplar, rice, and *Physcomitrella* genes in this large group as AAE (acyl activating enzyme-like) genes (Table S4).

A second group of adenylate-forming proteins (Fig. 1, highlighted by an oval) contained both bona fide 4CL proteins and previously annotated *Arabidopsis* 4CL-like (Costa *et al.*, 2003; Raes *et al.*, 2003; Shockey *et al.*, 2003; Ehling *et al.*, 2005) acyl-CoA synthetase (ACS) proteins. Strikingly, this ACS group is apparently land plant-specific. All angiosperm genomes, as well as the genome of the moss *Physcomitrella patens*, contained genes encoding proteins in this group, while no representatives from other eukaryote lineages were found. It is interesting, however, that *Penicillium* and

Dictyostelium contain genes relatively closely related to this land plant group. Inclusion of full-length gymnosperm *ACS* gene sequences, when they become available, in this analysis will allow definitive testing of the interpretation that this clade is restricted to land plants.

Within the apparent land plant-specific group, 5 clades containing angiosperm *ACS*s could be further delineated (Fig. 1; clades A–E). The *ACS* genes encoding proteins in these clades are phylogenetically closely related to *bona fide* 4CLs, which form a sister clade to clades A–E. As presented in more detail later, each of clades A–E contains at least one sequence representative of each of the four angiosperm plant species analysed (*Arabidopsis*, poplar, rice and maize), demonstrating that these proteins are evolutionarily conserved in the angiosperm lineage and that a common ancestor of these clades was present before the divergence of monocots and eudicots. The *Arabidopsis*, poplar, and rice proteins represented in the *bona fide* 4CL clade have been described and annotated (Ehlting *et al.*, 1999; Hamberger & Hahlbrock, 2004; Tsai *et al.*, 2006; Tuskan *et al.*, 2006; Hamberger *et al.*, 2007). The previously and currently annotated *Arabidopsis*, poplar, rice and *Physcomitrella* 4CL and *ACS* genes are given in Table 1.

Analysis of the *Physcomitrella* genome revealed four putative 4CL genes encoding proteins that grouped in the *bona fide* 4CL clade (Table 1; de Azevedo Souza *et al.*, unpublished), as well as five *ACS* genes falling outside the angiosperm *ACS* clades B–E. One *Physcomitrella* *ACS* sequence grouped into clade A (PpACS6; Table 1). This suggests that 4CL and clade A *ACS* genes originated early during the evolution of land plants before divergence of tracheophytes (vascular plants) and bryophytes, consistent with a possible role for these enzymes in the biosynthesis of phenylpropanoids and/or the extracellular matrix, a key innovation for adaptation to the land environment (Bowman *et al.*, 2007). Interestingly, *Physcomitrella* PpACS5 is basal to the *ACS* clades B–E, suggesting that it could represent an ancestral *ACS* gene retained in *Physcomitrella*. The remaining four *Physcomitrella* *ACS* sequences formed a distinct clade, suggesting bryophyte-specific evolution and diversification of this *ACS* gene family, which could encode proteins with functions distinct from those of angiosperm *ACS*s.

Almost all sequences in clades B, C, D and E, as well as *Physcomitrella* sequences PpACS1–4, contain a consensus PTS1 peroxisomal target sequence (Table 1, Fig. 2; Reumann *et al.*, 2004) at their C-termini, which suggests they are targeted to this organelle. Localization of *ACS* proteins to the peroxisome has been experimentally verified for *Arabidopsis* proteins in clade E (At4g05160, ACS6 and At5g63380, ACS9; Schneider *et al.*, 2005) and the *Arabidopsis* OPCL1 (At1g20510) protein in clade D (Koo *et al.*, 2006). Interestingly, all fungal *ACS*-related enzymes identified and shown in Fig. 1 also have peroxisomal target signals. Given that peroxisomes play a major role in fatty acid metabolism

in both plants and fungi, and that acyl:CoA ligases are widely used in modification of these molecules, it is possible that *ACS*s and 4CLs were recruited from fatty acid metabolism early in land plant evolution, to perform their current functions. None of the *ACS*s in clade A, most closely related to *bona fide* 4CLs, contained the PTS1 sequence, suggesting that loss of this sequence may have played a role in the acquisition of 4CL and subclade A functions. Furthermore, *Physcomitrella* PpACS5 located at a position basal to *ACS* clades B–D also lacks a PTS1 targeting sequence, suggesting a potentially distinct biochemical function for this enzyme, relative to the peroxisomally targeted *ACS*s.

The angiosperm-specific *ACS* clades were next analysed in more detail, focusing on complete gene families from *Arabidopsis*, poplar, and rice, for which whole genome sequence information is available. As shown in detail in Fig. 2 and Table 1, all three species contained *ACS* proteins in each of clades A–E. While the number of *ACS* genes within each genome was similar (13 in poplar, 12 in rice, and 9 in *Arabidopsis*; Table 1 and Fig. 2), the number in each clade varied between species. Certain *ACS* clades were greatly enriched for proteins from a particular species. For example, clade D is an *Arabidopsis* gene-rich clade, with five *Arabidopsis* representatives, two from poplar and only one rice member. By contrast, clade E is poplar rich with seven genes, three from rice and only one from *Arabidopsis*. Clade A, unique in containing *ACS* genes lacking the PTS1 targeting signal, is the only clade that contained a single representative from each species.

These data clearly show that *ACS* genes in different clades have undergone differential expansion in each angiosperm lineage, perhaps reflecting different events in the evolution of their genomes and differences in life histories that placed varying selective pressures on the elaboration of biochemical pathways requiring *ACS* activity. All four *Arabidopsis* *ACS* genes in clade D are located in tandem on chromosome 1, suggesting tandem duplication and selection for retention of the duplicated copies. Two of the poplar *ACS* genes in clade E (*PoptrACS11* and *PoptrACS12*) appear to have arisen by tandem duplication on linkage group XII. However, other members of this and other clades for which the poplar gene models are anchored to linkage groups are physically unlinked and on different linkage groups. This suggests that tandem gene duplication has not played a major role in diversification of the poplar *ACS* gene family. Many of the poplar *ACS* genes may rather have been retained after the salicoid whole genome duplication in the poplar lineage, in which chromosome doubling and subsequent rearrangement is thought to have increased the poplar chromosome complement from $n = 10$ to the current $n = 19$ (Tuskan *et al.*, 2006). For example, in the poplar-rich clade E, *PoptrACS10* and *PoptrACS11/12*, which are located on duplicated homoeologous linkage groups XII and XV, and *PoptrACS6* and *PoptrACS7*, which are located on duplicated homoeologous linkage groups XIII and X, are likely to have arisen in this manner (Tuskan *et al.*, 2006). Also noteworthy

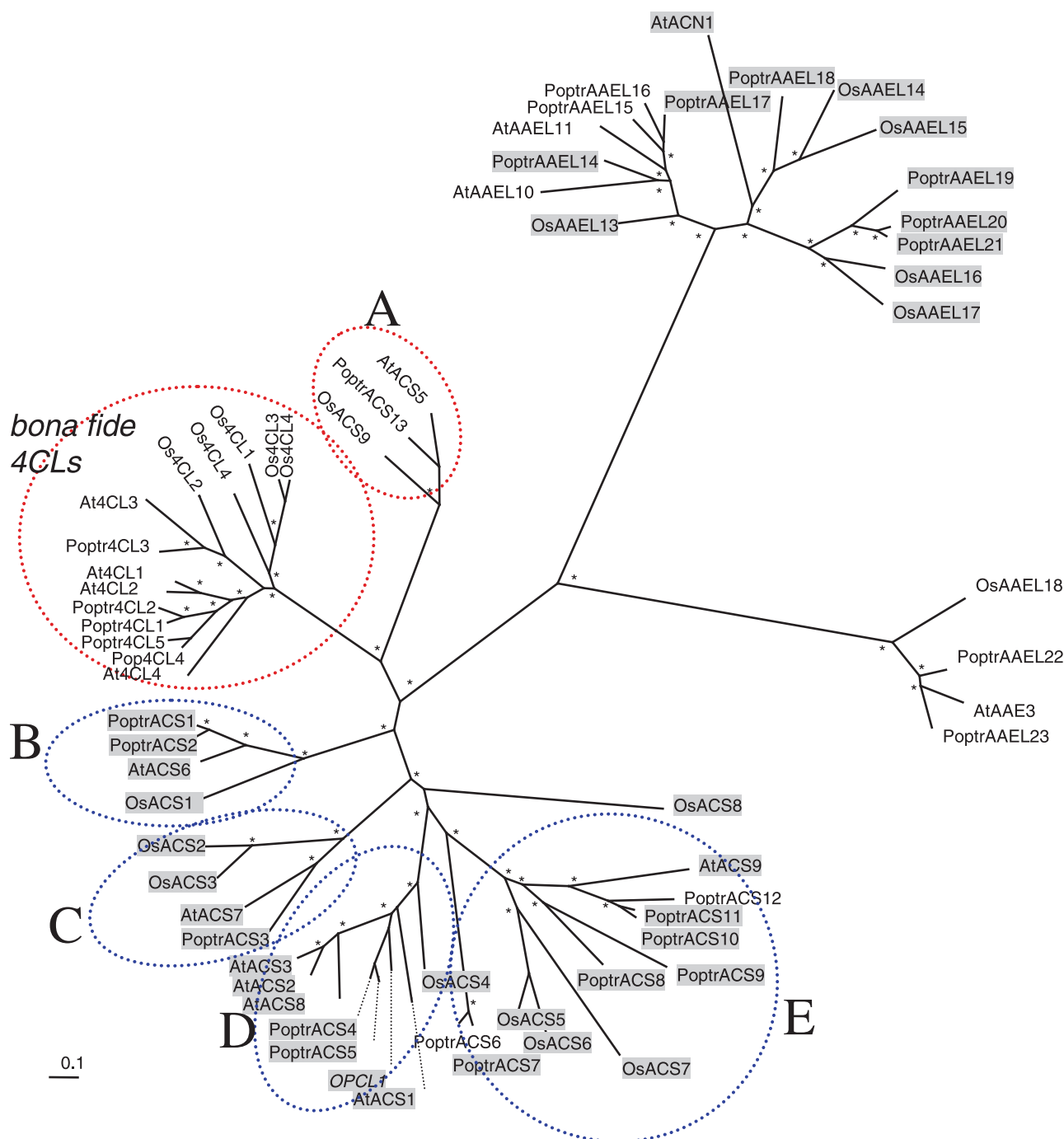


Fig. 2 Phylogenetic relationships of plant-specific acyl-CoA synthetases (ACSs) from three fully sequenced angiosperm genomes. Translated nucleotide sequences corresponding to ACS genes from *Arabidopsis*, poplar and rice were aligned and an unrooted phylogenetic tree generated. Nodes with bootstrap values above 70% are shown by stars. The 4-coumarate:CoA ligase (4CL) and ACS clades A–E discussed in the text are circled and contain at least one representative of each plant species. Protein names in shaded boxes contain the PTS1 peroxisomal target signal. Bar represents 0.1 amino acid changes.

among the poplar clade E ACS proteins is the loss of C-terminal PTS1 peroxisomal targeting sequences in two members (PoptrACS6 and PoptrACS12), suggesting that functional diversification may have taken place at the level of enzyme

localization in the poplar lineage after gene duplication. Taken together, these data show conservation of ACS gene number in all three lineages for some clades (A, B and C, with one or two members from each lineage), suggesting possible

conservation of function. Conversely, diversification of gene numbers in clades D and E has occurred in a lineage-specific manner, suggesting possible diversification of function in a lineage specific manner.

Developmental expression of poplar and *Arabidopsis* ACS genes

In order to gain insights into the possible functions of ACS proteins, we examined the developmental gene expression patterns of all *Arabidopsis* ACS genes, as well as representative poplar genes in clades A–E, by qRT-PCR. Expression profiles in different *Arabidopsis* and poplar organs and tissues are shown in Fig. 3, relative to the organ or tissue with the highest expression. To evaluate absolute expression levels, we also calculated the expression level of each gene tested relative to the respective *Arabidopsis* or poplar control gene in the organ or tissue where the gene was most highly expressed (Fig. 3). In general, *Arabidopsis* and poplar ACS genes from the same clade tended to have similar relative developmental expression patterns, and similar levels of expression. This is especially evident in those clades with single *Arabidopsis* and poplar ACS representatives. A striking example is clade A, in which *AtACS5* expression was strongly flower specific, and *PoptrACS13* had a similar pattern of flower-preferred expression. Interestingly, *PoptrACS13* expression was specific to male flowers, and a putative *AtACS5* orthologue in tobacco shows an anther-preferred expression pattern (Varbanova *et al.*, 2003), suggesting a role for ACS enzymes of this clade in a biochemical pathway important in anther and/or pollen development. Another example is the predominant expression of both *Arabidopsis* and poplar representatives of clade C in leaves, with lower expression in stem/xylem and flowers. Both the *Arabidopsis* and poplar representatives were expressed at low levels relative to the respective control genes, suggesting a potential specialized function (for example, expression in restricted cell types). Clade B contains two poplar genes and a single *Arabidopsis* member. Genes in this clade were expressed in all organs, but both poplar *PoptrACS2* and *AtACS6* showed highest expression in mature leaves, and lower expression in other organs. The pattern of *PoptrACS1* expression differed from that of *PoptrACS2*, with highest expression in flowers, bark and young leaves, suggesting subfunctionalization in expression patterns, as predicted from genes retained after gene duplication events (Duarte *et al.*, 2006). The *Arabidopsis* and poplar genes in this clade were expressed at similar, high levels relative to the control genes, but expression of *PoptrACS2*, the most highly expressed poplar ACS gene, was much higher than that of *PoptrACS1*, supporting the subfunctionalization of these genes at the level of expression.

More complex expression patterns were observed in clades where expansion of gene family members in either *Arabidopsis* or poplar has occurred. In clade D, the duplicated and highly

similar poplar genes *PoptrACS4* and *PoptrACS5* appeared to have similar expression patterns across a range of organs (Fig. 3), and shared similar, high expression levels. However, the transcribed portions of the two poplar genes were so similar that cross-detection cannot be excluded. By contrast, for the five representatives of the *Arabidopsis* members of clade D, distinct and complementary expression patterns were observed throughout the majority of organs tested, and the genes were expressed at remarkably different levels (0.51 relative to the control gene level for *AtASC2*, to 1482 relative to the control gene for *OPCL1*), which suggests expression subfunctionalization of these genes. The demonstration that the *Arabidopsis* *OPCL1* gene in this clade encodes an OPDA-CoA ligase (Koo *et al.*, 2006) is consistent with the very high developmental expression of *OPCL1* in flowers, where JA plays a role in anther development (Ito *et al.*, 2007). Given their phylogenetic relationships to *OPCL1*, other genes in this clade could share this activity, especially since some of them are wound and methyl jasmonate (MJ) inducible (Koo *et al.*, 2006 and later), and since the *opcl1* mutant retains the ability to make JA both developmentally and in response to wounding (Koo *et al.*, 2006). The distinct developmental expression patterns and expression levels of the *Arabidopsis* clade D ACS genes could therefore suggest specialization of at least some of the genes for developmental biosynthesis of JA in different organs.

The only *Arabidopsis* representative in clade E, *AtACS9*, was most highly expressed (at a high absolute level) in seedlings, followed by flowers. The three poplar homologues most closely related to *AtACS9* (*PoptrACS10*, *PoptrACS11* and *PoptrACS12*) displayed expression patterns largely complementary to each other, covering leaves, roots and male flowers, although all had high expression in roots. Like *AtACC9*, expression levels of the poplar genes were generally high, although they varied over 10-fold between each other. Thus, expansion of this family of ACS genes in poplar appears to have been accompanied by subfunctionalization for developmental expression.

Stress-induced expression of poplar and *Arabidopsis* ACS genes

The stress-responsiveness of the set of ACS genes, except the flower-specific genes in clade A (which are not expressed in vegetative organs in *Arabidopsis*), was tested at various times after mechanical wounding in both *Arabidopsis* and poplar. The ACS promoter-*GUS* fusions were constructed, using genomic sequences upstream of the ATG start codon, between 1.5 kb and 2 kb in length. Transgenic lines were generated, and promoter activity following wounding assayed in representative lines. Wound-induced transcription of *Arabidopsis* and poplar genes was also tested by qRT-PCR. In addition, expression of the poplar ACS genes was tested by qRT-PCR after each of the following treatments: herbivory by the forest

tent caterpillar (*Malacosoma disstria*; FTC), simulated herbivory (SH; wounding plus *Malacosoma disstria* regurgitant) and exposure to MJ.

Wound-induced expression data are shown in Fig. 4, and the responses of poplar genes are shown in Fig. 5. *Arabidopsis* 4CL2, known to be wound inducible (Ehrling *et al.*, 1999), was used as a positive control for *Arabidopsis* wound treatments, and was up-regulated over 5 fold at 4 h after wounding (data not shown). The endogenous *AtACS6* gene and *AtACS6* promoter-*GUS* transgene were not wound inducible (Fig. 4), and *PoptrACS2* in the same clade (B) was unresponsive both to wounding and other stress treatments (Figs 4 and 5). However, *PoptrACS1*, also in clade B, was upregulated 1.6-fold after 4 h wounding, and was strongly and transiently upregulated by SH and FTC treatments (Fig. 5). In a separate

microarray expression profiling experiment, *AtACS6* expression was not activated by diamondback moth herbivory (J. Ehrling and J. Bohlmann, unpublished). These data suggest that *AtACS6* and *PoptrACS2* do not function in wound or herbivory stress-related biochemical pathways, but that *PoptrACS1* may have gained this function after duplication of the gene in the poplar lineage, and that the biochemical pathway using the product of the enzyme encoded by *PoptrACS1* may play a role in defence against herbivory.

In clade C, *AtACS7* expression was downregulated by wounding to less than half the level of the unwounded control, and a similar result was obtained for the single poplar homologue in this group, *PoptrACS3*. Interestingly, *PoptrACS3* was strongly and transiently upregulated by SH but not other stresses. These results suggest that enzymes in this clade are

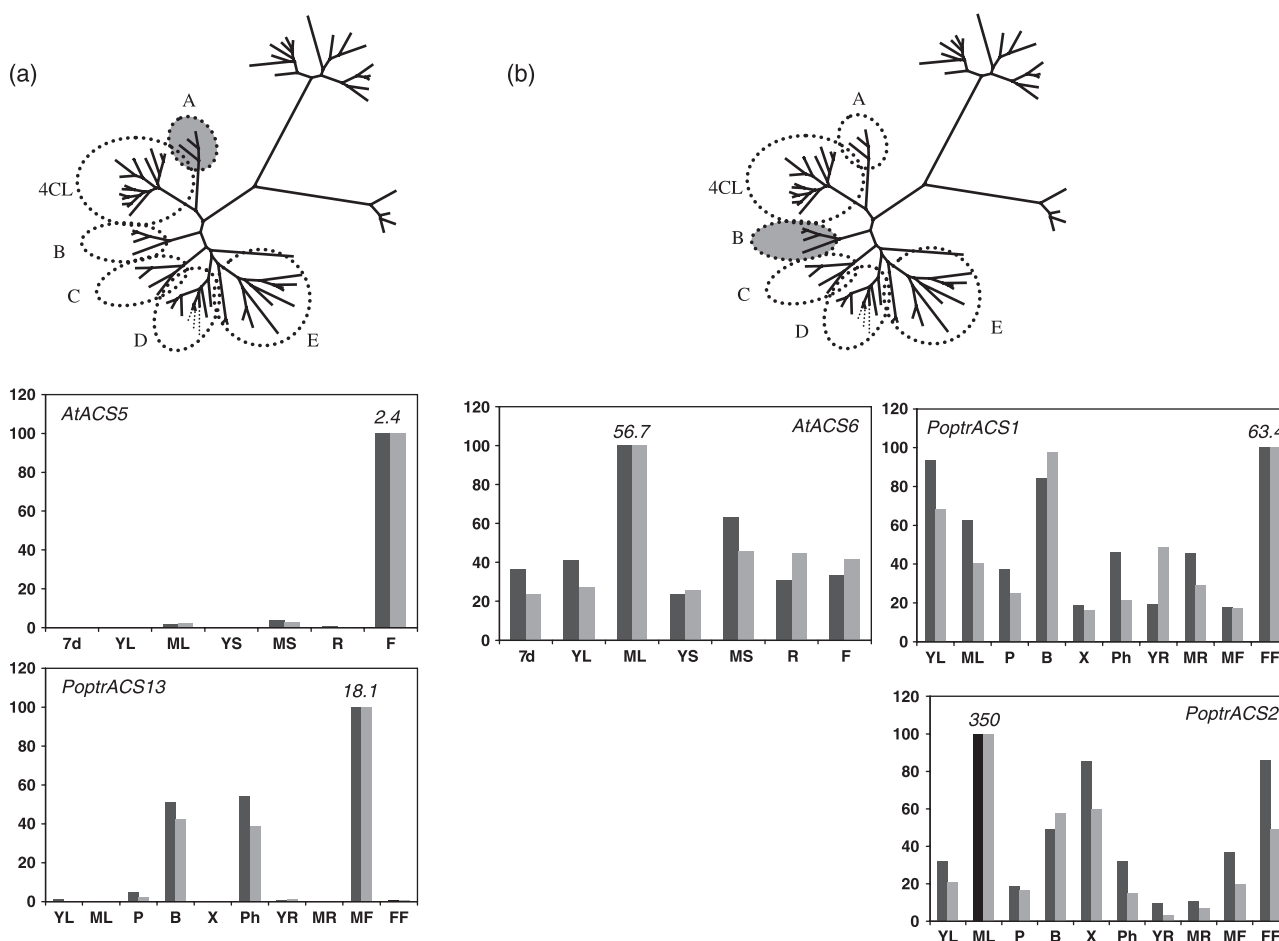


Fig. 3 Developmental expression profiles of ACS (acyl-CoA synthetase) genes in *Arabidopsis* and poplar. Expression levels were determined by quantitative reverse transcriptase polymerase chain reaction (qRT-PCR), and expression is represented relative to the tissue with the highest level of expression, set at 100%. Adjacent bars represent technical replicates using the same tissue samples. The level of expression of each gene relative to the *Arabidopsis* or poplar reference gene, set arbitrarily at 1.0, is given above the bars for the organs or tissue in which the gene was most highly expressed. Locations of ACS phylogenetic clades A–E for which expression data is shown are indicated by tinted circles. (a) Clade A genes; (b) Clade B genes; (c) Clade C genes; (d) clade D genes; (e) clade E genes. Abbreviations for *Arabidopsis* organs: 7 d, 7-d old seedlings; F, flower; ML, mature leaf; MS, mature shoot; R, root; YL, young leaf; YS, young stem. Abbreviations for poplar organs and tissues: B, bark; FF, female flower; MF, male flower; ML, mature leaf; MR, mature root; P, petiole; Ph, phloem; X, xylem; YL, young leaf; YR, young root.

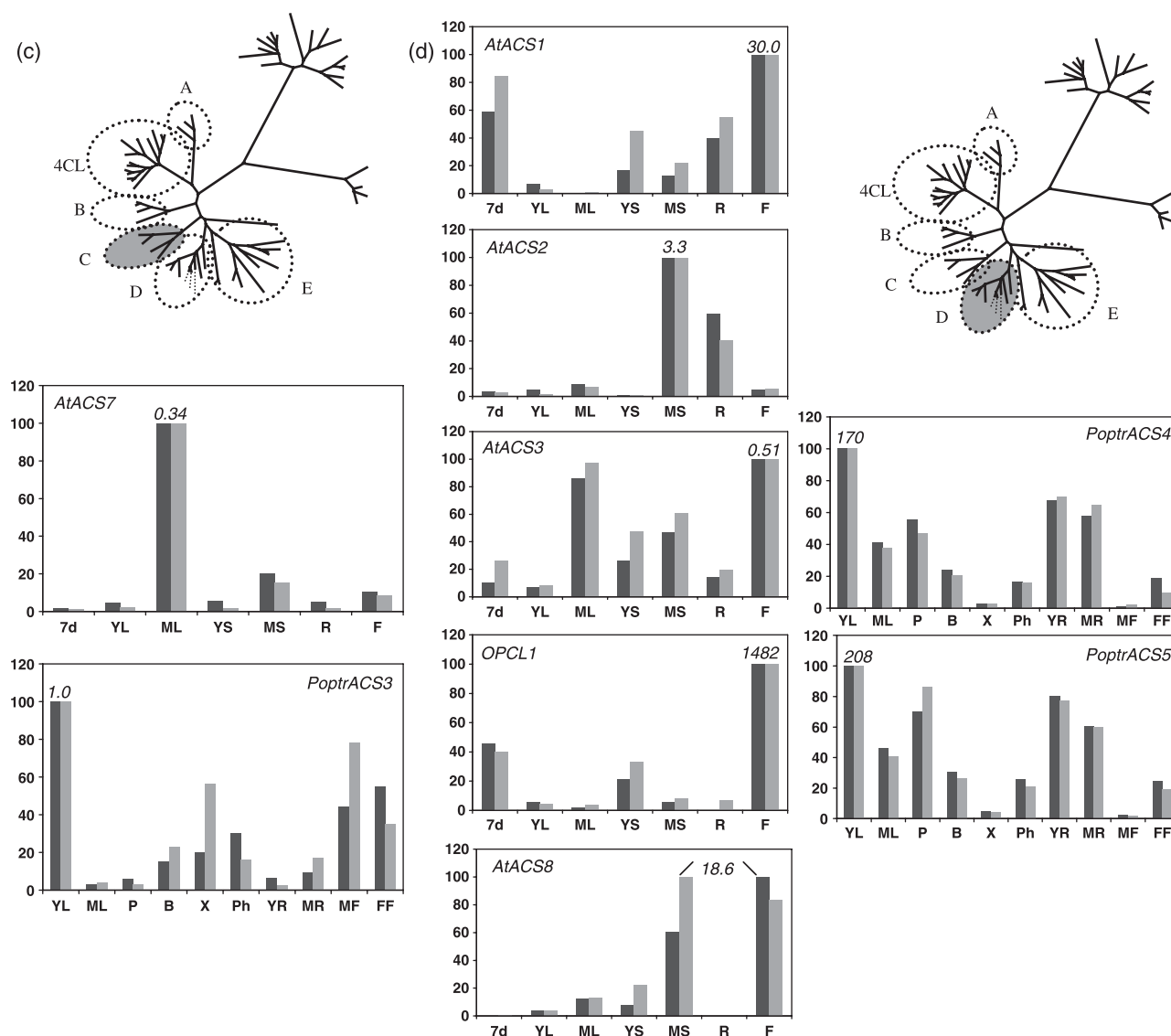


Fig. 3 continued

not likely to play roles in biochemical pathways related to herbivory and similar stresses, although the strong response of *PoptrACS3* to SH treatment warrants further investigation.

Arabidopsis and poplar genes in clade D were particularly responsive to wounding stress. Histochemical assays of transgenic promoter-*GUS* lines showed that the *Arabidopsis* *ACS2*, *ACS3* and *OPCL1* promoters were strongly responsive to wounding (Fig. 4a). Accumulation of *ACS2* and *ACS3* mRNA was also strongly upregulated by wounding, with *ACS3* levels 14-fold above the untreated the control within 1 h of wounding (Fig. 4b). While wound-induced accumulation of *OPCL1* and *AtACS8* mRNA was not obvious under the conditions used here, Koo *et al.* (2006) previously showed that expression of *OPCL1* is strongly, and *ACS8* weakly, wound inducible. *OPCL1* has been shown to encode a wound and MJ-inducible OPDA:CoA ligase involved in JA

biosynthesis (Koo *et al.*, 2006). Our data are consistent with potential roles of other *Arabidopsis* ACS genes in this clade in JA biosynthesis, in addition to *OPCL1*, perhaps as members of an *OPCL* gene family, as proposed by Koo *et al.* (2006). In support of this, *AtACS8*, as well as *OPCL*, is activated by diamondback moth herbivory based on microarray expression profiling (J. Ehrling and J. Bohlmann, unpublished data). Furthermore, analysis of public microarray gene expression data using the Bio-Array Resource, (<http://www.bar.utoronto.ca>; Toufighi *et al.*, 2005) showed that expression of *OPCL1*, *AtACS2* and *AtACS8* is induced by MJ treatment in 7-d-old seedlings. *AtACS3* is not represented in the microarray probe sets used in these experiments, but was strongly wound inducible in our experiments (Fig. 4). Unlike the other *Arabidopsis* members of Clade D, there is no evidence that *AtACS1* is wound or MJ inducible (Fig. 2), and it lacks

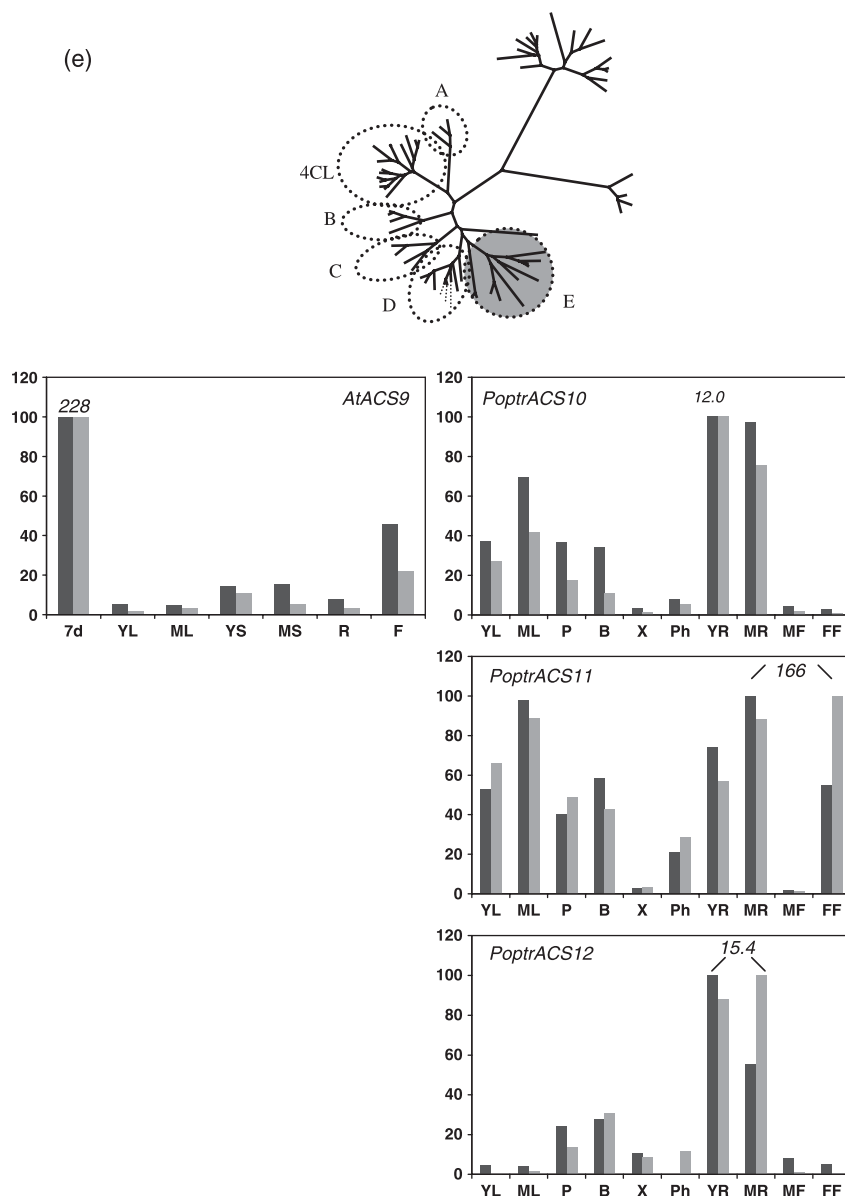


Fig. 3 continued

apparent activity against JA precursors *in vitro* (Kienow *et al.*, 2008) suggesting that it is less likely to be involved in JA biosynthesis. However, further analyses will be required to determine if *Arabidopsis* Clade D proteins other than AtOPCL1 have *in vivo* activities against OPDA, or preferentially use other substrates.

The expression of the poplar genes in this clade, *PoptrACS4* and *PoptrACS5*, was induced up to fivefold 4 h after wounding, and remained high 24 h after treatment (Fig. 4b). *PoptrACS4* and *PoptrACS5* were also strongly upregulated by herbivory, SH and MJ, with the last treatment leading to a transient 20-fold increase in expression by 2 h, with mRNA returning to control levels by 24 h (Fig. 5).

These two highly similar poplar genes are most similar to *Arabidopsis* *OPCL1* based on phylogenetic reconstruction (Fig. 2), and like *OPCL1* are highly expressed in a developmental context (Fig. 3d). These data are consistent with *PoptrACS4* and *PoptrACS5* encoding poplar OPDA:CoA ligases, a function that *OsASC4* (*Os03g04000*), the only rice gene in clade D (Fig. 2), could share. Interestingly, if this is the case, the two duplicated and highly similar poplar *OPCL* genes, and the single rice gene, contrast sharply in number to the expanded clade D family of potential *OPCLs* in *Arabidopsis*, suggesting that the regulation of JA biosynthesis in *Arabidopsis* may be more complex than in these other two angiosperm lineages.

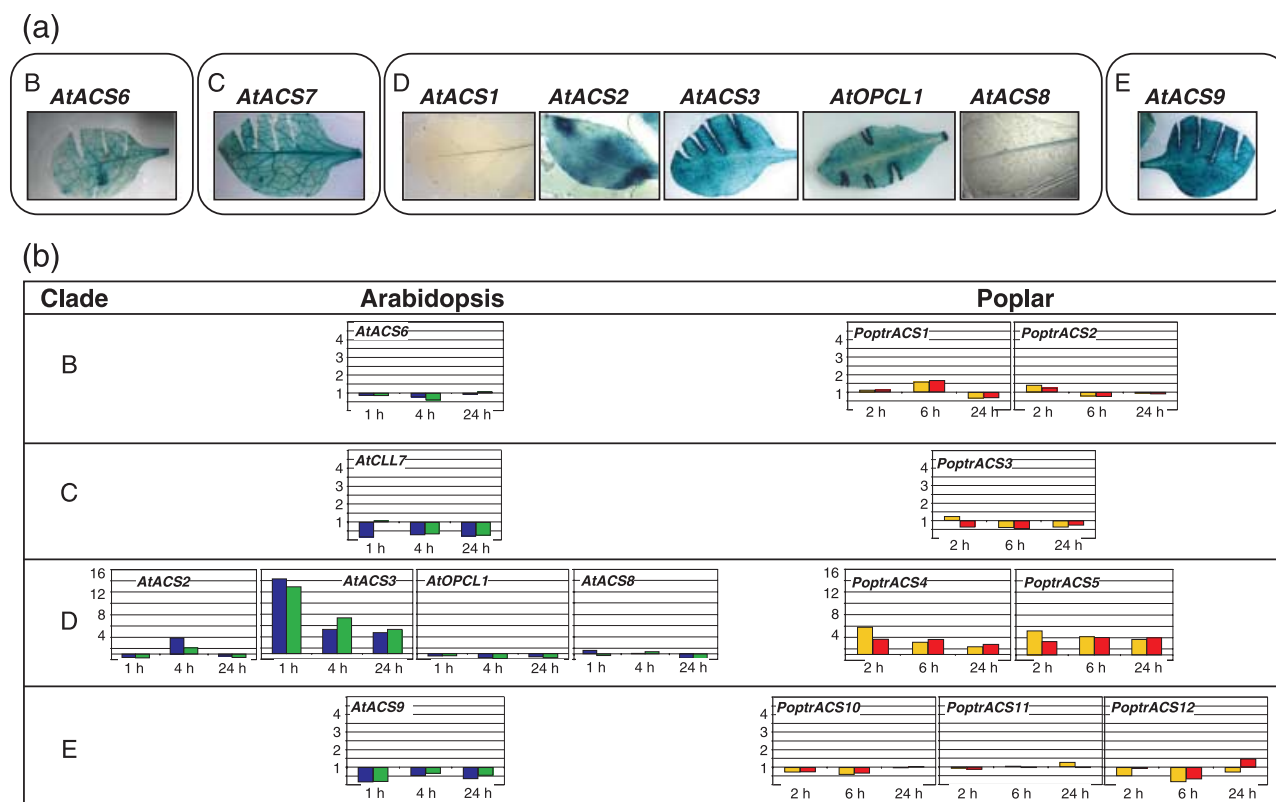


Fig. 4 Wound responsiveness of *Arabidopsis* and poplar peroxisomal ACS (acyl-CoA synthetase) genes in clades B–E. (a) Histochemical assay of glucuronidase (GUS) activity in transgenic *Arabidopsis* plants expressing ACS promoter–GUS gene fusions. Fully expanded rosette leaves were wounded 1 h before staining for GUS activity. (b) Quantitative reverse transcriptase polymerase chain reaction analysis of ACS gene expression in response to wounding in *Arabidopsis* and poplar. Data are expressed as fold-change in expression (y-axis) relative to unwounded control leaves. Adjacent bars represent technical replicates using the same tissue samples. Time (in hours) after wounding is given on the x-axis.

Expression of the single *Arabidopsis* clade E gene, *AtACS9* was downregulated after 1 h in response to wounding, and expression stayed below the control levels up to 24 h. Of the three poplar homologues most related to *AtACS9*, *PoptrACS12* showed the most similar expression pattern, whereas *PoptrACS10* and *PoptrACS11* showed little or no change in expression in response to wounding. The poplar genes in clade E had similar responses to FTC, SH, and MJ treatments (*PoptrACS12* was downregulated; *PoptrACS10* and *PoptrACS11* showed only minor fluctuations). The enzyme encoded by *AtACL9* has been shown to be a fatty acyl:CoA synthetase, with activity *in vitro* with fatty acids and OPDA, a precursor in JA biosynthesis (Schneider *et al.*, 2005; Kienow *et al.*, 2008). It is localized to the peroxisome and has been suggested to be a potential OPDA:CoA ligase (Schneider *et al.*, 2005; Kienow *et al.*, 2008). However, the lack of wound, herbivory and MJ activation of *AtACL9* and of the most closely related poplar ACS genes, as well as the phylogenetic distance between this gene and clade D containing the *Arabidopsis OPCL1* gene, do not support a role for these enzymes in the stress induced synthesis of JA, and the *in vivo* substrate of *AtACL9* remains to be clarified.

The clade E gene *PoptrACS12* has distinct expression patterns relative to the most closely related genes in poplar, *PoptrACS10* and *PoptrACS11*. It shows downregulated expression in response to wounding and related stresses (Figs 4 and 5), and shows a contrasting developmental expression pattern, with highest expression in roots, phloem and bark (Fig. 3e). Interestingly, the *PoptrACS12* protein lacks the PTS1 peroxisomal targeting sequence (Fig. 2) suggesting, that while *PoptrACS10* and *PoptrACS11* may have enzymatic functions similar to the *AtACS9* protein as peroxisomally localized fatty acyl:CoA synthetases, *PoptrACS12* may have acquired a new function specific to the poplar lineage, as a nonperoxisomal CoA ligase.

Subcellular localization of *PoptrACS5*

We hypothesized that *PoptrACS5* is an orthologue of the *Arabidopsis OPCL1* gene. To test this, we attempted to express *PoptrACS5* heterologously in *Escherichia coli*, but were unable to generate a strain expressing sufficient *PoptrACS5* protein to test its activity *in vitro*. To test the subcellular localization of *PoptrACS5*, we generated an N-terminally tagged

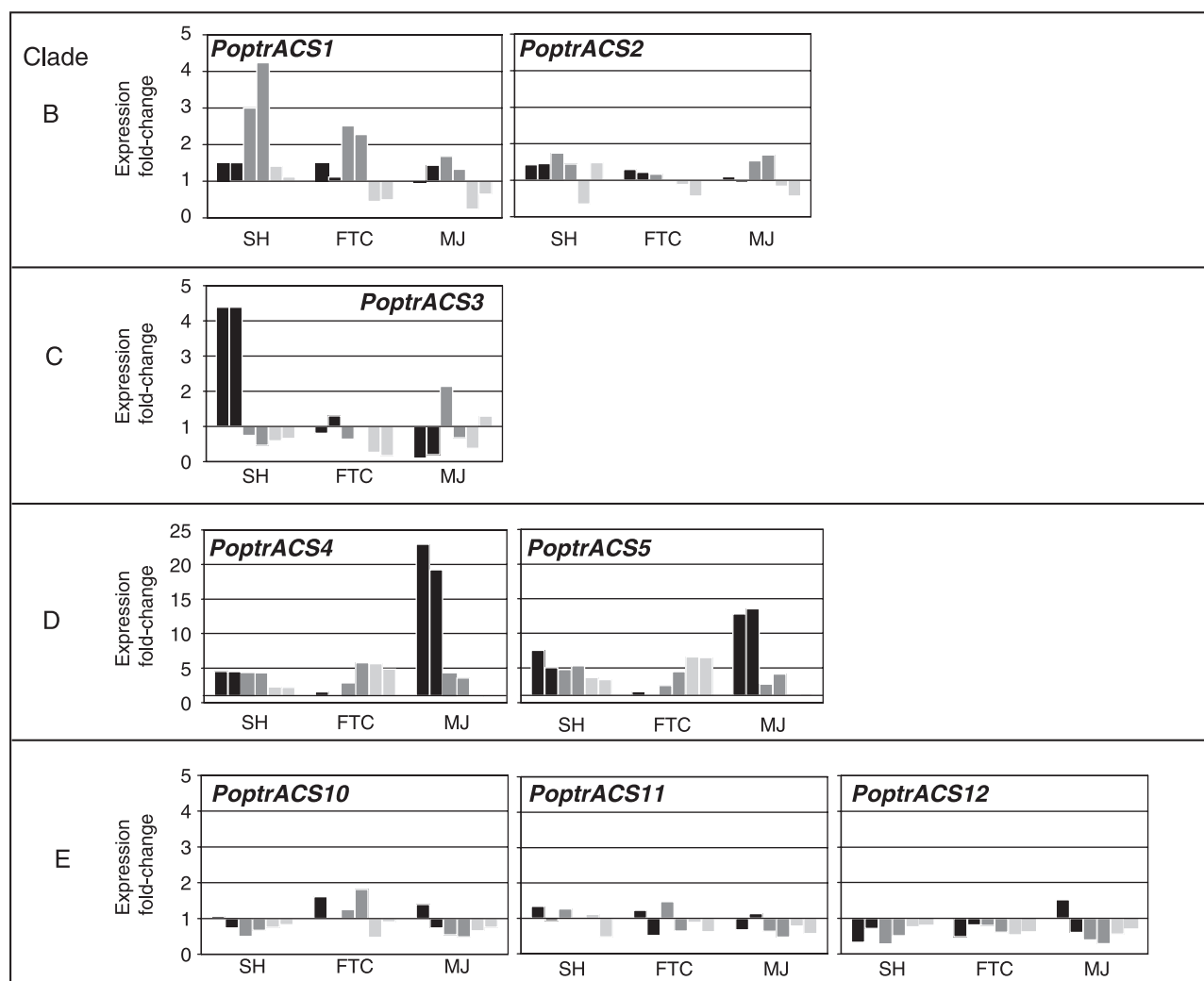


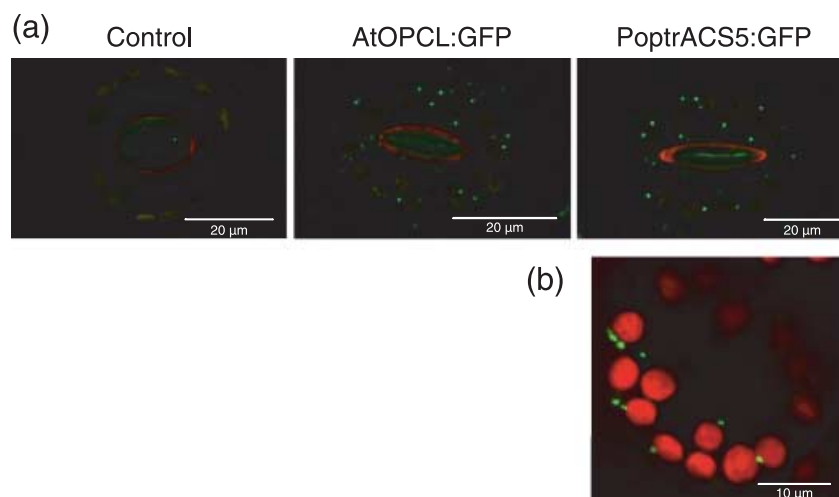
Fig. 5 Stress responsiveness of poplar ACS (acyl-CoA synthetase) genes in clades B–E. Quantitative reverse transcriptase polymerase chain reaction analyses were carried out on RNA isolated from poplar leaves after simulated herbivory (SH, mechanical wounding plus insect regurgitant), herbivory by the forest tent caterpillar (*Malacosoma disstria*; FTC), and exposure to methyl jasmonate (MJ) for 2 h (closed bars), 6 h (dark tinted bars), and 24 h (light tinted bars). Data are expressed as fold-change in expression relative to untreated control leaves or leaves sprayed with the solvent Tween20, in the case of MJ treatment. Adjacent bars of the same shading represent duplicate determinations using the same tissue samples. The reference gene used was that encoding elongation initiation factor 5A (GenBank accession CV251327).

PoptrACS5 and *Arabidopsis* OPCL1-GPF fusions, generated transgenic tobacco lines, and assayed reporter fluorescence in mesophyll and epidermal cells (Fig. 6). This analysis showed that both proteins were localized to similar subcellular structures consistent with a peroxisomal localization, as previously demonstrated by Koo *et al.* (2006) for AtOPCL1. Mesophyll cell peroxisomes in which PoptrACS5-GFP accumulated appeared to be physically associated with chloroplasts (Fig. 6b). This is consistent with the necessity of shuttling intermediates in JA biosynthesis between the plastids, where OPDA is produced, and peroxisomes, where OPDA:CoA thioesters are generated in preparation for acyl chain shortening to produce JA (Li *et al.*, 2005; Koo *et al.*, 2006).

Conclusions

Our data show that the 4CL-like ACS enzymes are a conserved, land plant-specific group of adenylate-forming enzymes most closely related to true 4CL enzymes among a larger group of plant and microbial acyl activating and acyl:CoA synthetase enzymes. The fact that ACS representatives, as well as true 4CL enzymes, are found in the moss *Physcomitrella* suggests that both groups of enzymes evolved early in the transition of plants to the terrestrial environment and were present in the most recent common ancestor of bryophytes and vascular plants. The ACS enzymes are largely predicted to be peroxisomally located and are related to peroxisomal enzymes found in fungi and prokaryotes. This suggests that

Fig. 6 Subcellular localization of AtOPCL and the poplar homologue PoptrACS5 in guard cells of transgenic tobacco lines. (a) Green fluorescent protein (GFP) fluorescence lines expressing *N*-terminal AtOPCL:GFP and PoptrACS5:GFP proteins. An untransformed line was used as negative control. Yellow signals derive from chlorophyll autofluorescence and GFP fluorescence is green. (b) Detailed view of PoptrACS5:GFP showing close proximity of peroxisomes (showing GFP fluorescence) with chloroplasts (red autofluorescence).



the progenitors of land plant 4CL and the related ACS enzymes may have originally performed peroxisomal functions and that bona fide 4CL enzymes and clade A ACS enzymes evolved from such progenitor enzymes by loss of the PTS1 peroxisomal targeting sequence and acquisition of new functions in other cellular compartments. It appears that loss of peroxisomal targeting sequences can occur relatively easily in this group of enzymes, since two apparent independent examples in the duplicated poplar *ACS* genes (*PoptrACS6* and *PoptrACS12*, both in clade E), and the *Physcomitrella* *PpACS5* gene were found. The nonperoxisomal poplar proteins have presumably acquired new functions in the poplar lineage, subsequent to the gene duplication events.

The *in vivo* biological functions of the angiosperm ACS enzymes are for the most part still unknown, although such functions may be conserved within the conserved angiosperm clades A–E, and in many cases they have expression patterns that suggest functions in developmental and/or stress-related biochemical pathways not related to phenylpropanoid metabolism. In addition to *Arabidopsis* OPCL1 and its poplar homologues, other stress-induced ACS enzymes in clade D may have acyl-activating functions associated with chain-shortening reactions in JA biosynthesis. AtACS6 (At4g05160; clade B) and AtACS9 (At5g63380; clade E) have activity against a variety of acyl substrates including OPDA or its derivatives and long-chain fatty acids (Schneider *et al.*, 2005; Kienow *et al.*, 2008), although *in vivo* substrates have not been identified. Thus, many or all of the ACS proteins may accept a diversity of different fatty acid or other acyl substrates. Additional data from surveys of potential substrates (Schneider *et al.*, 2005; Kienow *et al.*, 2008), 4CL and ACS structural information, and *in vivo* functional assays (for example using reverse genetic approaches) will be necessary to determine functions of *Arabidopsis* ACL enzymes.

We targeted *Arabidopsis* ACS genes in two of the clades, A and E, with single *Arabidopsis* representatives (*AtACS5* and

AtACS9, respectively, Fig. 2) for reverse genetic analysis using T-DNA insertion null-mutant lines (data not shown). While no metabolic or morphological phenotypes were evident for a homozygous *acs9* mutant line (data not shown), the *acs5* mutant showed a striking male sterility phenotype (data not shown; C. de Azevedo Souza, C. Douglas *et al.*, unpublished). This is consistent with roles for *AtACS5* and the poplar homologue *PoptrACS13* in anther development predicted from their developmental expression patterns (Fig. 3a). Further comparative biochemical studies on enzymes encoded by the corresponding poplar, rice, and *Physcomitrella* ACS genes will help to shed light on the conservation of pathways utilizing ACS enzymes in land plant lineages.

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Supporting Information

Additional supporting information may be found in the online version of this article.

Table S1 Primers used for quantitative reverse transcriptase polymerase chain reaction

Table S2 Primers used to clone *Arabidopsis ACS* promoters for fusion to the *GUS* reporter gene

Table S3 Polymerase chain reaction primers used to generate green fluorescent protein (GFP) fusions

Table S4 Gene names and sequences used in Fig. 1

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